

4.3 Connections with Cohomology

The Hurewicz theorem provides a strong link between homotopy groups and homology, and hence also an indirect relation with cohomology. But there is a more direct connection with cohomology of a quite different sort. We will show that for every CW complex X there is a natural bijection between $H^n(X; G)$ and the set $\langle X, K(G, n) \rangle$ of basepoint-preserving homotopy classes of maps from X to a $K(G, n)$. We will also define a natural group structure on $\langle X, K(G, n) \rangle$ that makes the bijection a group isomorphism. The mere fact that there is any connection at all between cohomology and homotopy classes of maps is the first surprise here, and the second is that Eilenberg–MacLane spaces are involved, since their definition is entirely in terms of homotopy groups, which on the face of it have nothing to do with cohomology.

After proving this basic isomorphism $H^n(X; G) \approx \langle X, K(G, n) \rangle$ and describing a few of its immediate applications, the later parts of this section aim toward a further study of Postnikov towers, which were introduced briefly in §4.1. These provide a general theoretical method for realizing an arbitrary CW complex as a sort of twisted product of Eilenberg–MacLane spaces, up to homotopy equivalence. The most geometric interpretation of the phrase ‘twisted product’ is the notion of fiber bundle introduced in the previous section, but here we need the more homotopy-theoretic notion of a *fibration*, so before we begin the discussion of Postnikov towers we first take a few pages to present some basic constructions and results about fibrations.

As we shall see, Postnikov towers can be expressed as sequences of fibrations with fibers Eilenberg–MacLane spaces, so we can again expect close connections with cohomology. One such connection is provided by *k-invariants*, which describe, at least in principle, how Postnikov towers for a broad class of spaces are determined by a sequence of cohomology classes. Another application of these ideas, described at the end of the section, is a technique for factoring basic extension and lifting problems in homotopy theory into a sequence of smaller problems whose solutions are equivalent to the vanishing of certain cohomology classes. This technique goes under the somewhat grandiose title of Obstruction Theory, though it is really quite a simple idea when expressed in terms of Postnikov towers.

The Homotopy Construction of Cohomology

The main result of this subsection is the following fundamental relationship between singular cohomology and Eilenberg–MacLane spaces:

Theorem 4.57. *There are natural bijections $T : \langle X, K(G, n) \rangle \rightarrow H^n(X; G)$ for all CW complexes X and all $n > 0$, with G any abelian group. Such a T has the form $T([f]) = f^*(\alpha)$ for a certain distinguished class $\alpha \in H^n(K(G, n); G)$.*